

জুনিয়র ইনস্ট্রাক্টর চাকরি সিলেবাস অনুসারে



SOFTMAX
ONLINE SCHOOL

CST

LIVE
CLASS

ডাটা স্ট্রাকচার এন্ড অ্যালগরিদম

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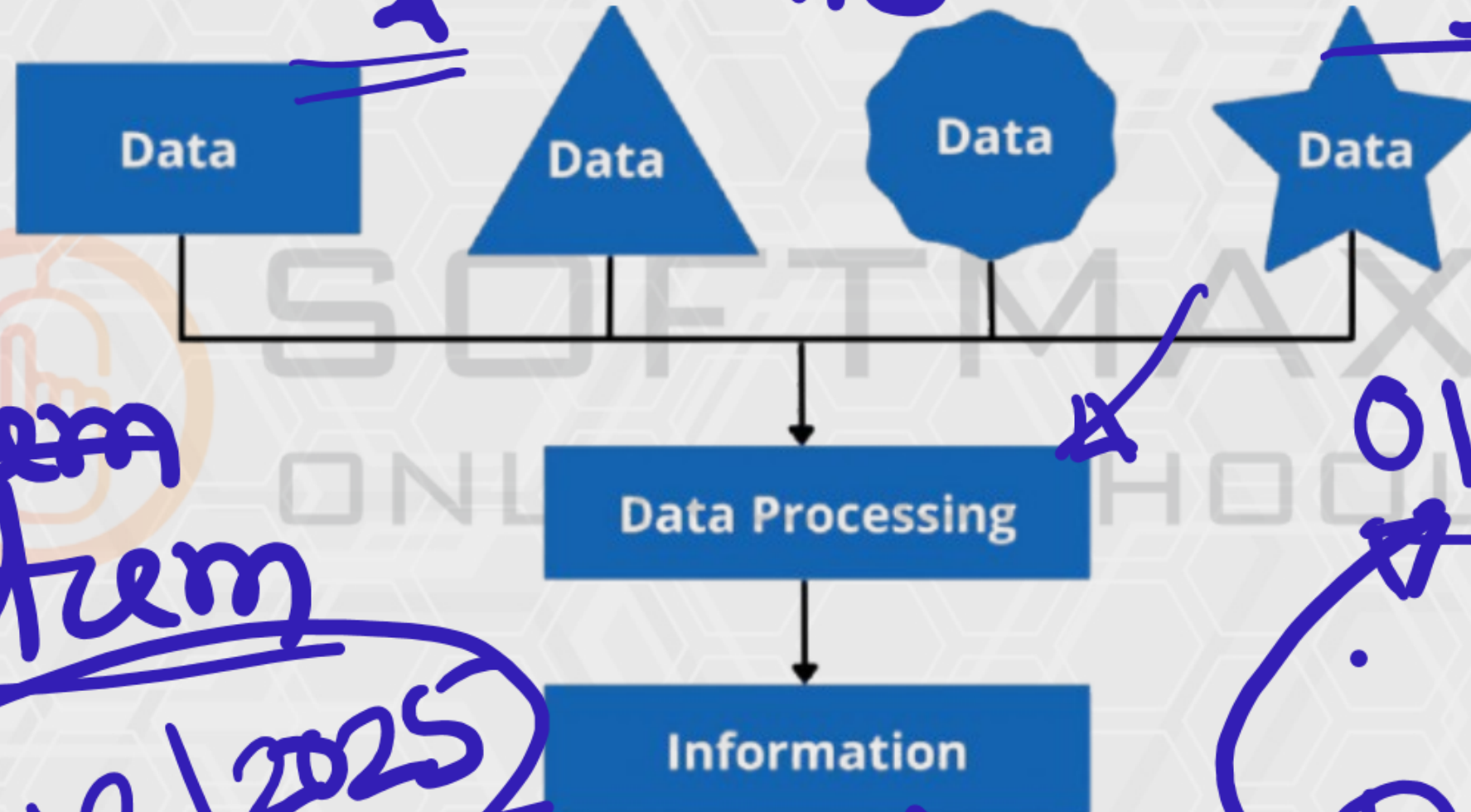




Syllabus

- 
- 
1. Data Structure Basic
 2. Array
 3. Linked List
 4. Stack
 5. Queue
 6. Searching
 7. Sorting

Data & Information



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ତଥ୍ୟ

~~Data~~
Datum

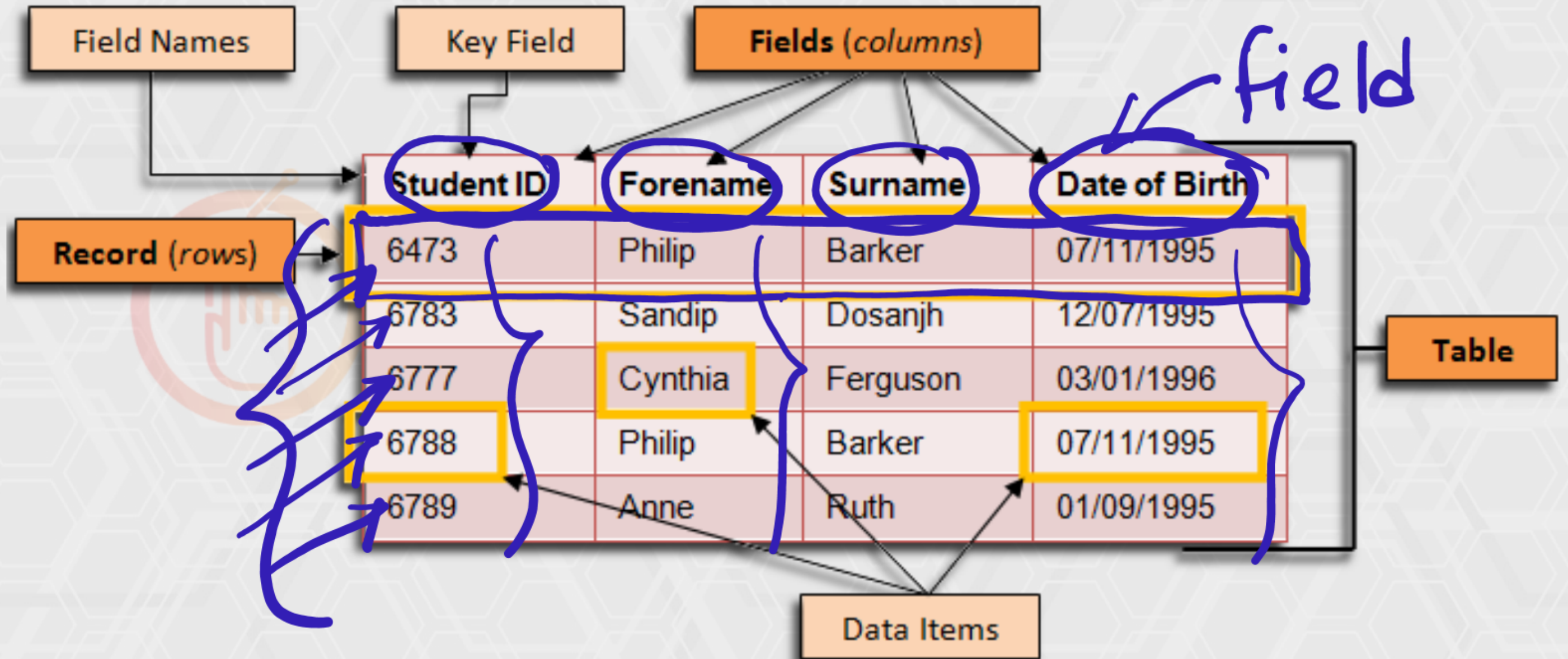
30/12/2025

01712.....

Rubel

Rubel

Field, Record & File





Data Structure

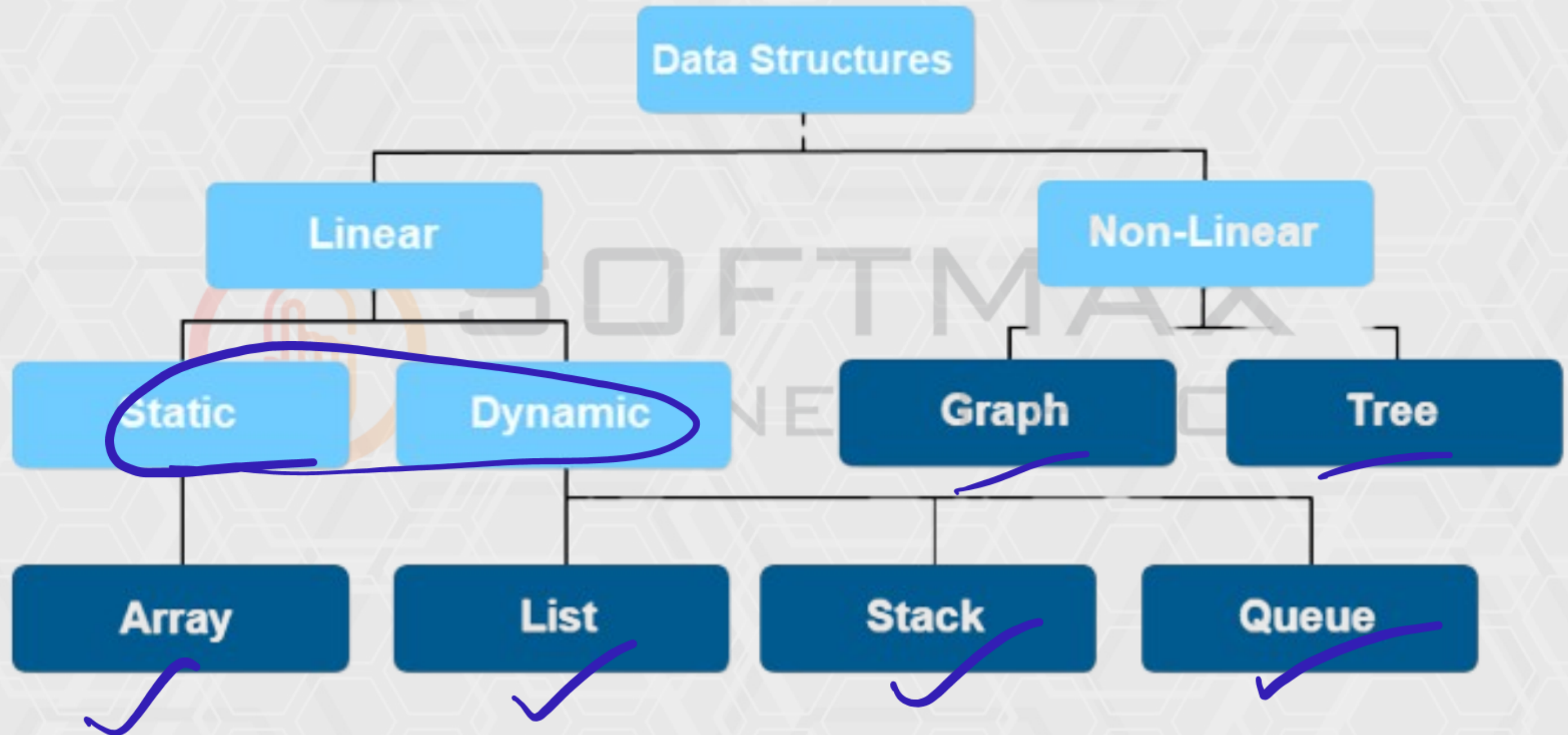


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দ্বিগুণ উদ্যোগে
চাকরি যুদ্ধে...

Types of Data Structure



দ্বিগুণ উদ্যোগে
চাকরি যুদ্ধে...



Linear Data Structure

LINEAR

→ সার্বসুক্রমিক

Arrays

Stacks

Linked
Lists

Queues



99 100 101 102



Non Linear Data Structure

NON
LINEAR

Trees

Graphs

Tables

Sets

দ্বিগুণ উদ্যোগে
চাকরি যুদ্ধে...

Operation of Data Structure



Insertion



Deletion



Searching



Traversing



Sorting

Merging

Ascending

A-Z
0-9

Descending

Z-A
9-0

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চাকরি যুদ্ধে...





Algorithm & Flowchart



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চাকরি যুদ্ধে...

Symbol of Flowchart

Symbol	Meaning
	Start/Stop
	Process
	<u>Input/Output</u>
	Decision/Branching
	<u>Connector</u>
	Flow

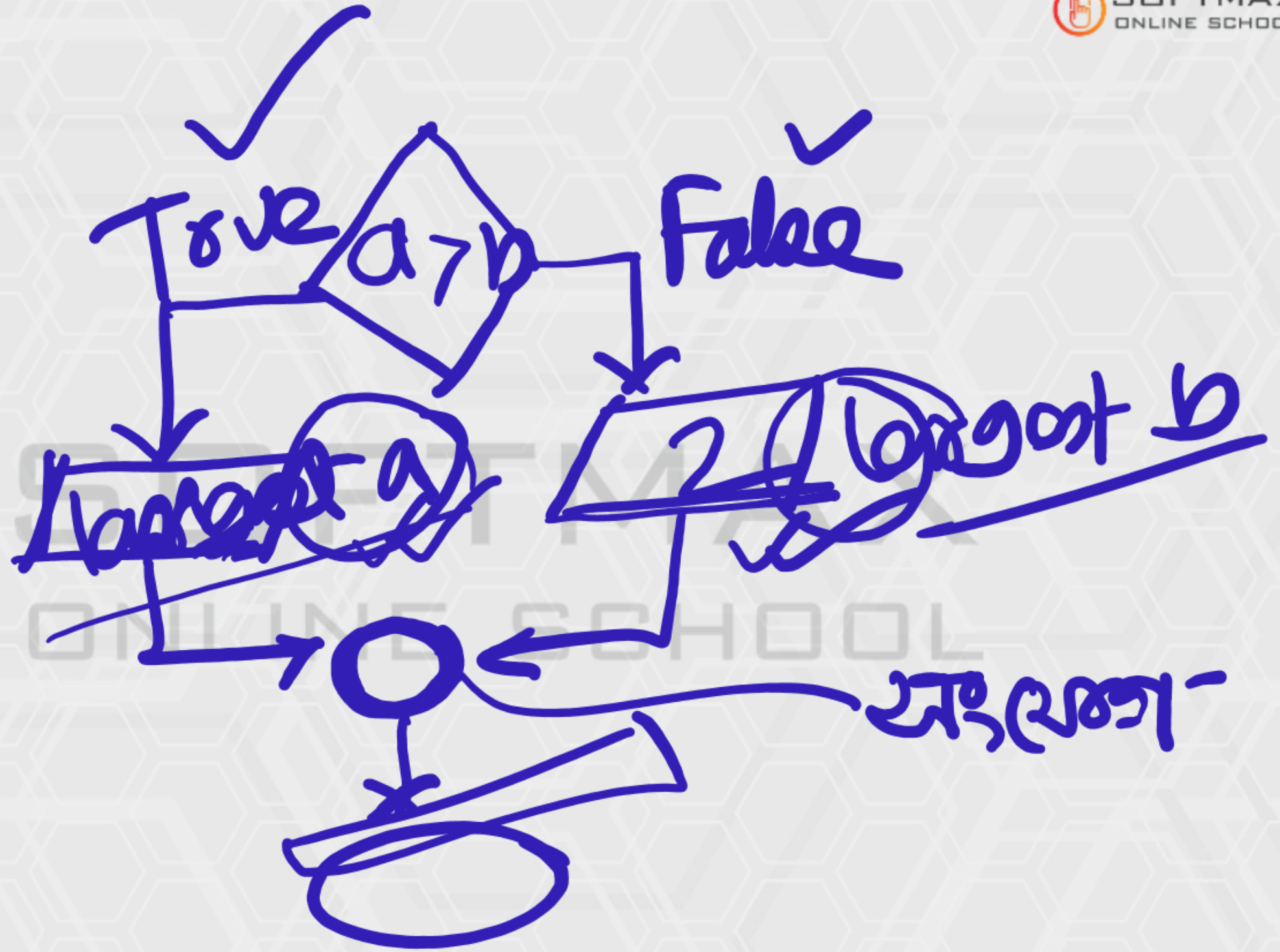
প্রবাহ



দ্বিগুণ উদ্যোগে
চাকরি যুদ্ধে...

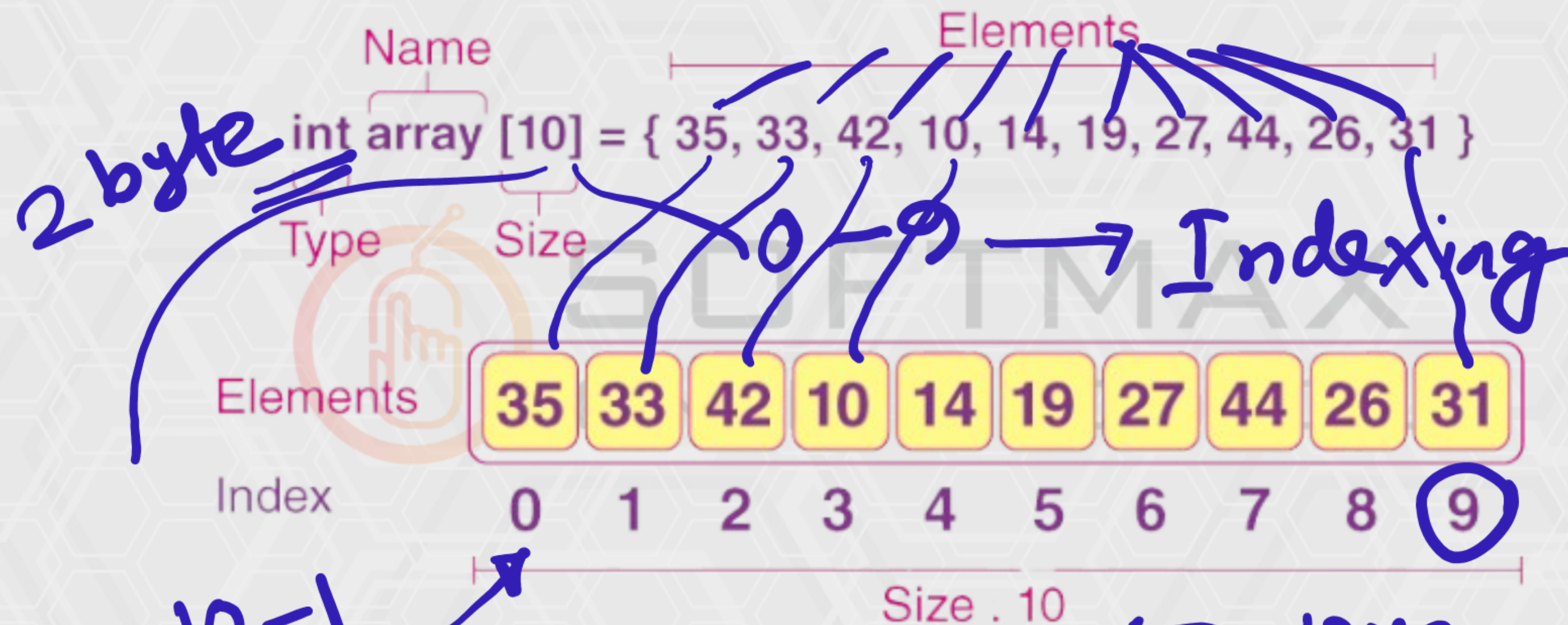


$a = 30$
 $b = 20$





Array Data Structure



`int a[10];` $10 \times 2 = 20 \text{ byte}$
 $= 20 \times 8 \text{ bit}$
 $= 160 \text{ bit}$

দ্বিগুণ উদ্যমে
সকরি যুদ্ধে...



int a, b, c, d, ...

Syntax: `int a[100];` 2 byte
4 byte

Data type Array Name [size];

Statement

int
char
float
double

←→

দ্বিগুণ উদ্যোগে
সফটম্যাক্স

Types of Array

On the basis of
Size

Fixed
Size

`int a[10];`

2D

Dynamic
Size

`int a[];`

3D

On the basis of
Dimensions

One-
Dimensional
Array

1D

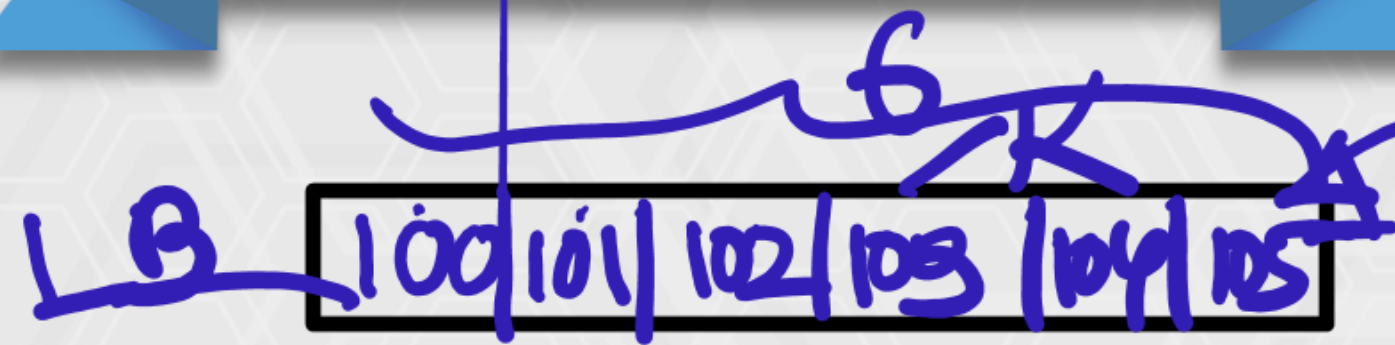
Multi-
Dimensional
Array

Two-Dimensional Array

Three-Dimensional Array

... and so on

One Dimensional Array



UB = Upper Bound

LB = Lower u

✓ Element সংখ্যা = $UB - LB + 1$
 $= 105 - 100 + 1 = 6$

✓ $A[loc[K]] = BA + w(K - LB)$

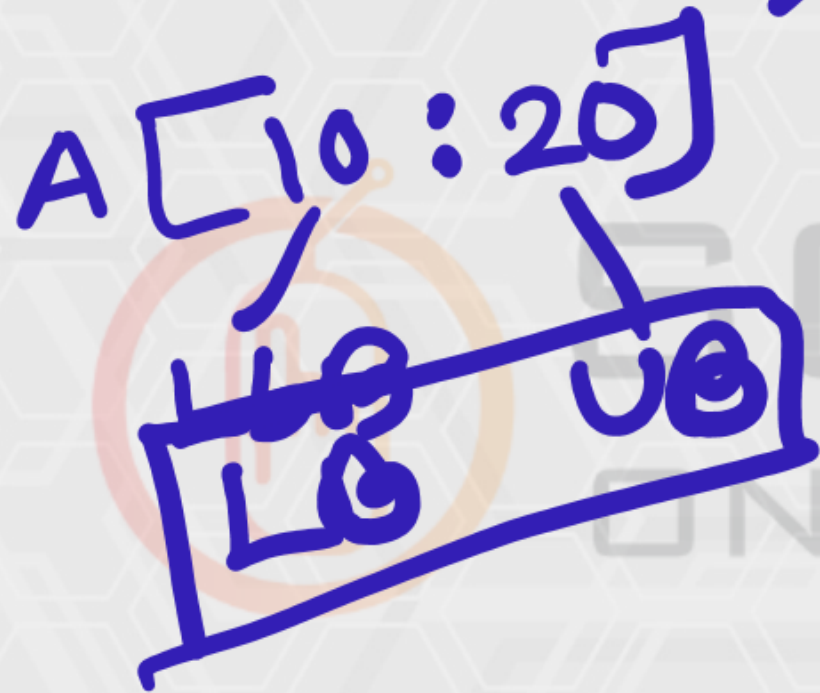
BA = Base Address = Starting Address.

W = word size

K = অর ইন্ডেক্স

$$A(\text{LOC}[K]) = BA + W(K - LB)$$

$$= 1000 + 4(12 - 10)$$



$$= 1000 + (4 \times 2)$$

$$= 1008$$

1000

④



1000 1004 1008

Two Dimensional Array

int a [3] [3];

Row
Column

	0	1	2	
0	(0,0)	(0,1)	(0,2)	Column Index
1	(1,0)	(1,1)	(1,2)	
2	(2,0)	(2,1)	(2,2)	
	Row Index			

$$3 \times 3 = 9$$

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চাকরি যুদ্ধে...

Multi Dimensional Array

~~row~~
int a[r][c]

Subscript
int a[2][3][3]

Second
Dimension

Third
Dimension

First
Dimension



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চাকরি যুদ্ধে...



0

$$\frac{3 \times 3}{9}$$

2, 3, 3

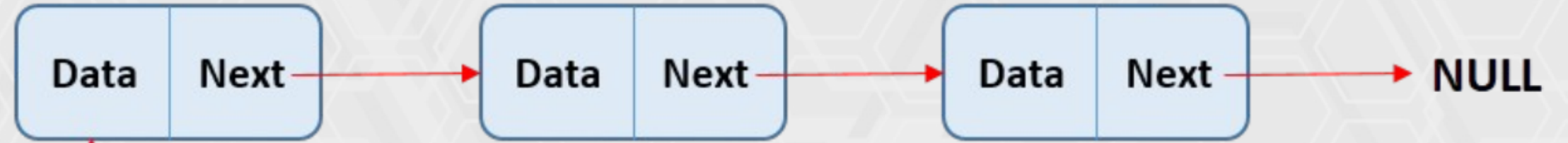
$$2 \times 3 \times 3 = 18$$

1

$$\frac{3 \times 3}{9}$$



Linked List



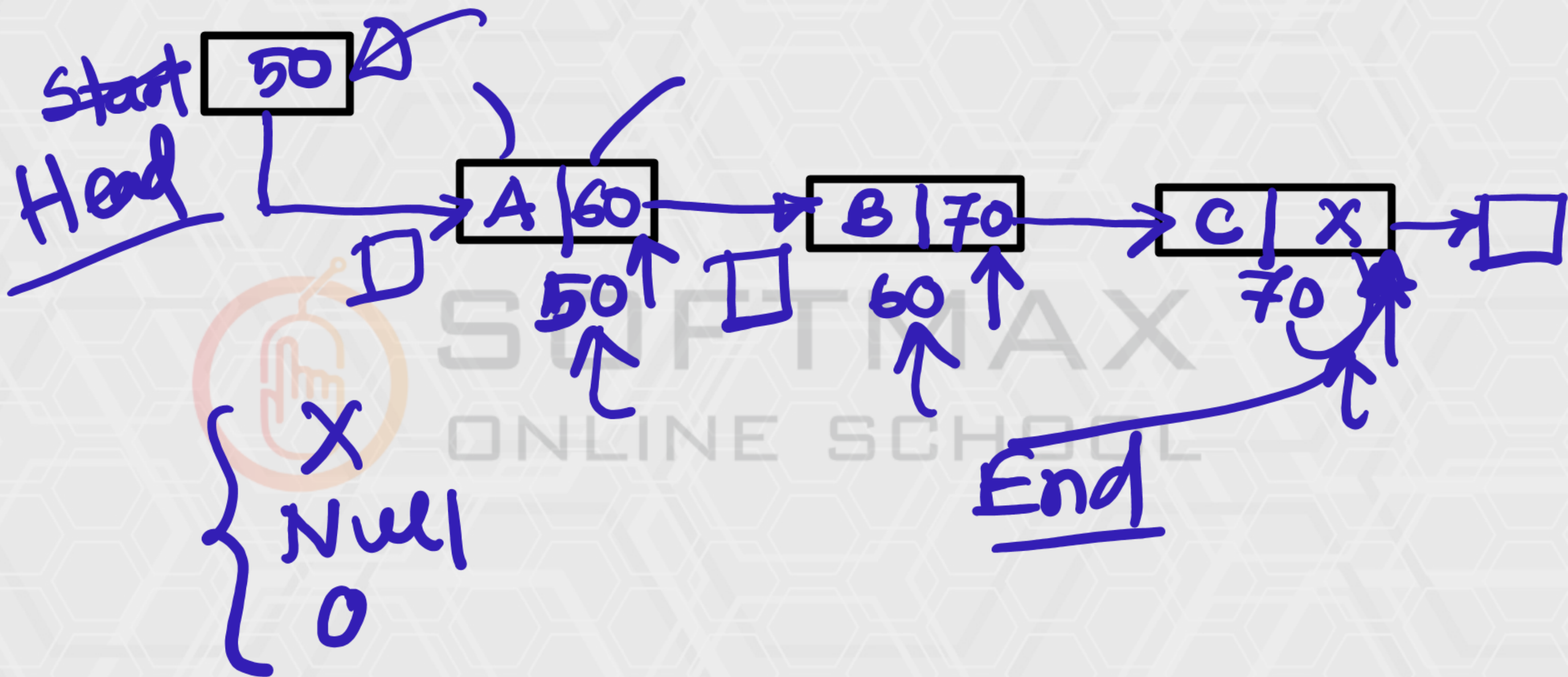
HEAD

Node

Data / NPA
info

Next pointer Address.
 $100 \times 2 = 200 \text{ byte}$
 $10 \times 2 = 20 \text{ byte}$
12



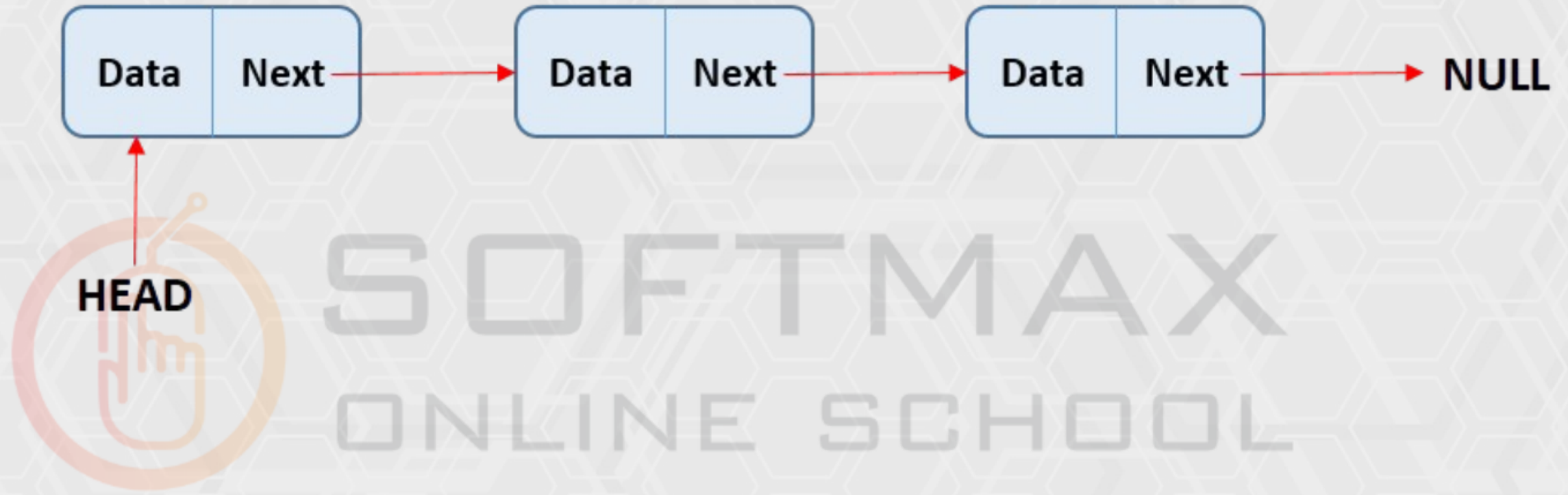




Types of Linked List

1. Linear
2. Doubly ✓✓✓✓
3. Circular
4. Header.

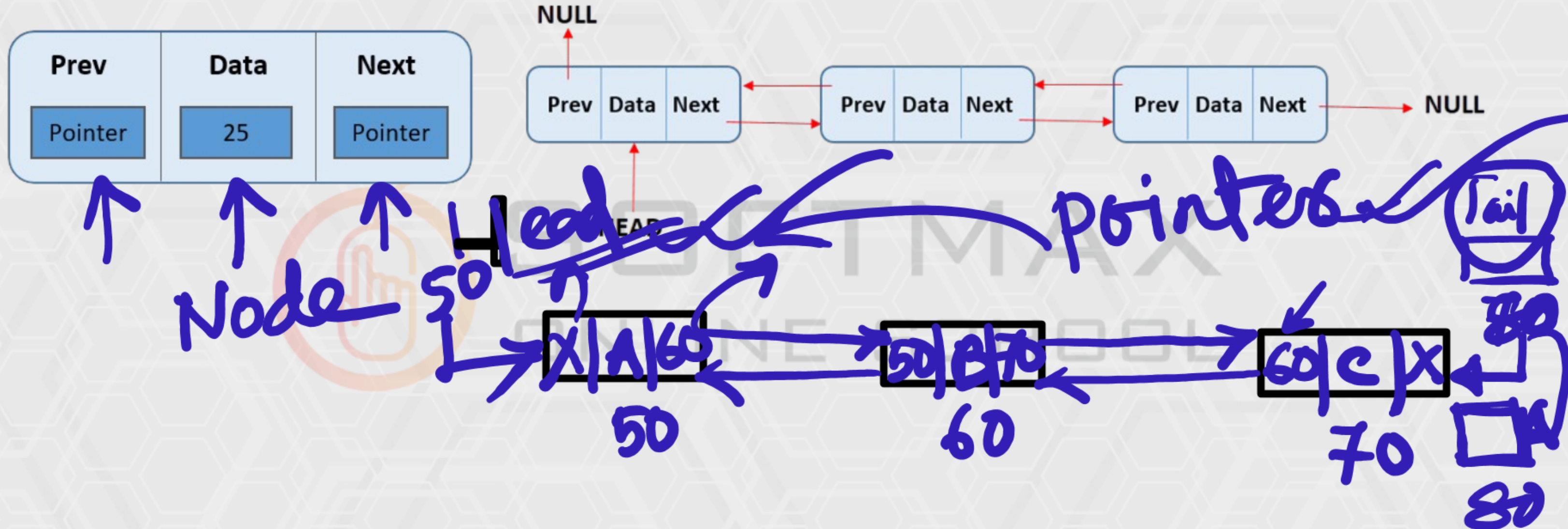
Linear Linked List



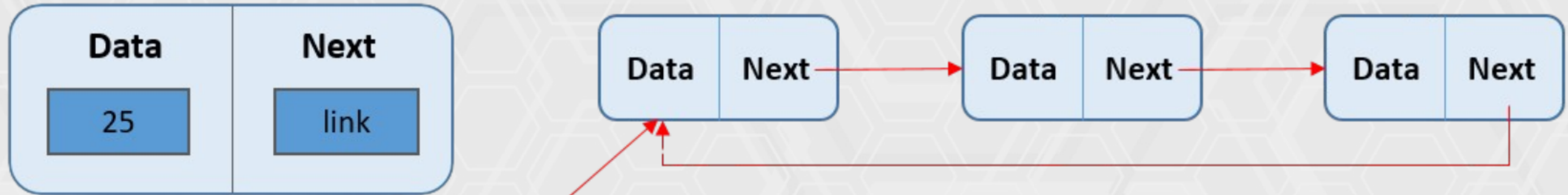
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চাকরি যুদ্ধে...



Doubly Linked List



Circular Linked List



HEAD

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চাকরি যুদ্ধে...

Header Linked List

